

TikTok's Discovery Commerce and the Power of Invisible AI Infrastructure

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“When all you see is what you love, how could it not be fun?” The slogan from TikTok’s advertising campaign captures the essence of its AI-powered recommendation engine. In September 2023, TikTok expanded that capability with the launch of **TikTok Shop**, introducing a new shopping paradigm known as **Discovery Commerce**. Rather than searching for products directly, users discover items unexpectedly while scrolling. As entertainment seamlessly evolves into purchasing, TikTok has become more than a social platform—it has reshaped the entire consumer journey.

TikTok Shop generated approximately **US\$33.2 billion (KRW 48 trillion)** in gross merchandise value (GMV) in 2024 and is projected to surpass **US\$66 billion (KRW 96 trillion)** in 2025. Across its **1.6 billion global users**, including **170 million in the United States**, consumers are embracing a new shopping experience in which products are discovered unexpectedly while scrolling through content. Live shopping hosted by creators transforms products into engaging entertainment, while shoppers become creators themselves by sharing reviews under hashtags such as **#TikTokMadeMeBuyIt**, fueling another cycle of consumer engagement.

Behind this experience lies a powerful AI infrastructure. At **Oracle AI World**, Oracle introduced **Project Texas**, offering a glimpse into the infrastructure that supports TikTok’s recommendation engine. After facing the possibility of a U.S. ban in 2020 over concerns surrounding data privacy and national security, TikTok partnered with Oracle to migrate U.S. user data into a dedicated Oracle Cloud region. This move addressed data sovereignty and security concerns while providing a stable foundation for operating within the U.S. market.

Securely storing data, however, is only part of the equation. To analyze the real-time behavioral data generated by hundreds of millions of users and instantly deliver AI-powered recommendations, TikTok also requires ultra-high-speed networking and GPU-based AI infrastructure.

Understanding the preferences of more than **150 million users** requires collecting enormous volumes of real-time behavioral signals—including clicks, swipes, comments, and watch time—alongside the continuous stream of video metadata generated every second. To support this workload, TikTok connects edge data centers to Oracle Cloud’s central AI training region through a **400 Gbps FastConnect** dedicated network. The company also reduced the number of network hops from data collection to processing, minimizing latency across the system.

The scale of this infrastructure is remarkable. A **400 Gbps** connection delivers substantially greater bandwidth than both a typical **1 Gbps** home internet connection and many competing **100 Gbps** dedicated enterprise networks. Data transmitted at these extraordinary speeds is immediately available for AI training and inference. The moment a user taps “Like,” TikTok’s AI is already calculating which video should appear next. By combining ultra-high-speed networking with GPU-powered AI infrastructure, the platform can deliver hyper-personalized recommendations in as little as **0.1 seconds**.

Equally impressive is the efficiency of TikTok's AI computing architecture. The platform leverages Oracle Cloud's **RoCE (RDMA over Converged Ethernet)**-based GPU superclusters to maximize AI inference performance. Originally developed to accelerate Oracle database workloads, RDMA technology has evolved into a core component of modern AI infrastructure. Thousands of GPUs exchange data directly through **Direct Memory Access (DMA)** without passing through CPUs or operating systems, enabling AI models to train and perform inference at extraordinary speed. Like high-performance sports cars racing down a traffic-free **400 Gbps highway**, the system analyzes user preferences almost instantaneously and recommends content with the highest likelihood of driving engagement and purchases.

TikTok's success is not simply a triumph of recommendation algorithms. It is a triumph of engineering. While users immerse themselves in shopping experiences, **400 Gbps FastConnect** serves as the platform's digital expressway, and **RoCE-powered GPU clusters** continuously perform real-time AI inference behind the scenes. In the end, **Discovery Commerce** is made possible by infrastructure that remains almost entirely invisible to the user.

The lesson for Korea's fashion industry is equally clear. Trend-driven products and engaging content alone are no longer enough to compete on the global stage. Sustainable competitive advantage will depend on AI infrastructure capable of processing global-scale traffic while uncovering the hidden preferences of consumers around the world. In the age of AI, infrastructure is just as important as content.